



Kinetic Energy and Work

KE

Work

Power

Energy and Kinetic Energy

- Energy is a scalar quantity
- Associated with the state of an object/system
- Units: joule (J)
- **Kinetic Energy K** - associated with the state of motion of an object.

$$K = \frac{1}{2}mv^2$$

- $1\text{J} = 1\text{kg}\cdot\text{m}^2\text{s}^{-2}$
- For atoms, electrons, ions
 - 1 electron volt = $1\text{ eV} = 1.6 \times 10^{-16}\text{J}$
 - 1keV
 - 1MeV
 - 1GeV

Work

- **Work W** is energy transferred to or from an object by means of a force acting on the object
- Sign convention:
 - Energy transferred to the object is +ve
 - Energy transferred from the object is -ve
- Units are same as energy – J
- No material is transferred (e.g. electronic transfer)
- Example: Push hard against a wall – NO WORK IS DONE.
No energy is transferred to the wall.
Get tired due to muscles contracting.
- For this chapter – W represents work, weight represented by mg

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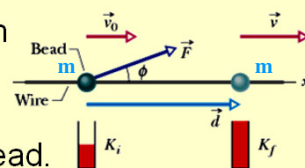
Work and Kinetic Energy

- Consider a bead which can slide on a frictionless wire (along x axis).
- Constant force $\vec{F}$ applied at an angle ϕ to the wire – accelerates bead.
- $F_x = ma_x$
- Moves through displacement $\vec{d}$
- Velocity changes from $\vec{v}_0$ to $\vec{v}$

$$v^2 = v_0^2 + 2a_x d$$
- Solve for a_x and substitute in $F_x = ma_x$

$$\frac{1}{2}mv^2 - \frac{1}{2}mv_0^2 = F_x d$$

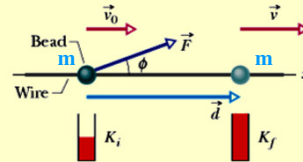
$$K_f - K_i = F_x d$$
- Kinetic energy has been changed by the force – Work done $W = F_x d$



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Work and Kinetic Energy

- Kinetic energy has been changed by the force – Work done $W = F_x d$
- $F_x = F \cos \phi$
- $W = Fd \cos \phi$
- $W = \vec{F} \cdot \vec{d}$
- F is the magnitude of $\vec{F}$ and $\vec{d}$ is the displacement
- Restrictions to using these equations:
 - Force must be constant
 - Object must be particle-like i.e. rigid, all parts move together in same direction.
- Work will be positive if $\phi < 90^\circ$, Work will be negative if $180^\circ > \phi > 90^\circ$



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Work and Kinetic Energy

- Units for Work: $1\text{J} = 1\text{kg}\cdot\text{m}^2\text{s}^{-2} = 1\text{ N}\cdot\text{m}$
- Work done by multiple forces
 - Net work done on the object
 - Sum of the work done by individual forces.
 - Either
 - Find work done by each force and add these works
 - Or
 - Determine the net force on the object and then determine the work done by net force $\vec{F}_{net}$ if ϕ is the angle between the net force and displacement $\vec{d}$
 - **Work-Kinetic Energy Theorem**

$$\Delta K = K_f - K_i = W_1 + W_2 + W_3 + \dots$$

$$= W_{net}$$

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Work – Kinetic Energy Theorem

- $\Delta K = K_f - K_i = W_1 + W_2 + W_3 + \dots$
 $= W_{net}$
- $\left(\begin{array}{c} \text{Change in KE} \\ \text{of particle} \end{array} \right) = \left(\begin{array}{c} \text{Net work done} \\ \text{on particle} \end{array} \right)$
- $K_f = K_i + W$
- $\left(\begin{array}{c} \text{KE after net work} \\ \text{done on particle} \end{array} \right) = \left(\begin{array}{c} \text{KE before net} \\ \text{work done} \end{array} \right) + \left(\begin{array}{c} \text{Net work} \\ \text{done} \end{array} \right)$

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Checkpoint 1

- A particle moves along an x axis. Does the kinetic energy of the particle increase, decrease or remain the same if the particle's velocity changes
 - From -3 m/s to -2 m/s
 - From -2m/s to 2 m/s
- In each situation is the work done on the particle positive, negative or zero?

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Problem 9

- The only force acting on a 2.0 kg canister that is moving in an xy plane has a magnitude of 5.0 N. The canister initially has a velocity of 4.0 m/s in the positive x direction and some time later has a velocity of 6.0 m/s in the positive y direction. How much work is done on the canister by the 5.0 N force during this time?

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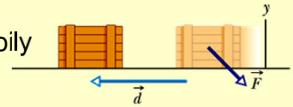
Problem 13

- A luge and its rider, with a total mass of 85 kg, emerge from a downhill track onto a horizontal straight track with an initial speed of 37 m/s. If a force slows them to a stop at a constant rate of 2.0 m/s^2
 - What is the magnitude F of the net force required?
 - What distance d do they travel while slowing down, and
 - What work W is done on them (luge and rider) by the force?

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Sample Problem p146

- During a storm, a crate is sliding across a slick oily parking lot through a displacement $\vec{d} = (-3.0 \text{ m})\hat{i}$ while a steady wind pushes against the crate with a force $\vec{F} = (2.0 \text{ N})\hat{i} + (-6.0 \text{ N})\hat{j}$. The situation and coordinate axes are shown in the Figure.



- How much work does this force do on the crate during the displacement?
- If the crate has a kinetic energy of 10 J at the beginning, what is its kinetic energy at the end of the displacement?

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Work done by some specific forces

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Work done by Gravitational force / weight

- A tomato with a mass m is thrown upwards
- Initial speed - v_0 , $K_i = \frac{1}{2}mv_0^2$
- Gravitational force causes the speed to decrease, KE decreases due to W done (by F_g).
- Work done during this displacement $\vec{d}$:
- $W = Fd \cos \theta$

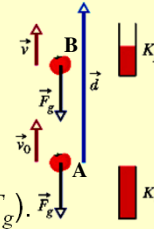
$$= mgd \cos \theta = mgd \cos 180^\circ$$

$$= -mgd$$

- When the tomato falls back down:

$$W = mgd \cos \theta = mgd \cos 0^\circ$$

$$= mgd$$



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Work done by Gravitational force / weight

- Work done in lifting and lowering an object.
- Force $\vec{F}$ is applied to lift an object.
- Displacement is upwards.
- Applied force does positive work.
- Gravitational force does negative work.
- Energy is transferred **to** the object by $\vec{F}$
- Energy is transferred **from** the object by $\vec{W} \left(\vec{F}_g \right)$.



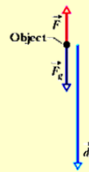
$$\Delta K = K_f - K_i = W_F + W_{mg}$$

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Work done by Gravitational force / weight

- Lowering the object:
- Energy is transferred **from** the object by $\vec{F}$
- Energy is transferred **to** the object by $\vec{W}$ ($\vec{F}_g$).

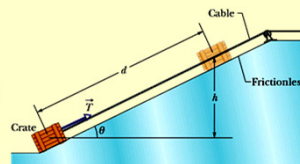
$$\Delta K = K_f - K_i = W_F + W_{mg}$$



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Problem

- A 15.0 kg crate, initially at rest, is pulled, via a cable, a distance $d = 5.70$ m up a frictionless ramp to a height $h = 2.50$ m, where it stops.



- How much work is done on the crate by the gravitational force during the lift?
- How much work is done on the crate by the force T during the lift?

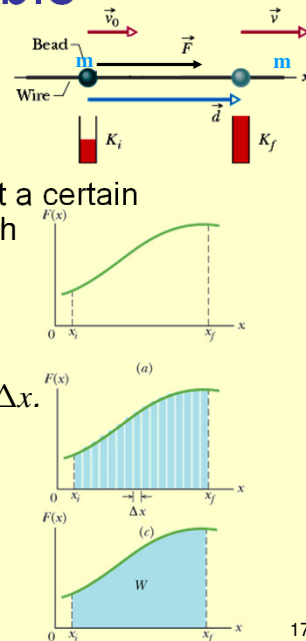
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Work done by a variable force

- One dimension:
- Force is in the positive x direction and the magnitude varies with position x . At a certain position, the force does not change with time.
- The graph shows a 1-D variable force.
- Can't use $W = Fd \cos \phi$.
- Divide the displacement into intervals Δx .
- If Δx is small, the average value of the force can be taken as constant.
- In the limit $\Delta x \rightarrow 0$

Integration

$$W = \int_{x_i}^{x_f} F(x) dx$$



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Work done by a variable force

- Three dimensions
- Consider a 3-D force acting on a particle.
- $\vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$
- F_x, F_y, F_z on the position of the particle
- F_x is dependent on x , but not on y or z ;
 F_y is dependent on y , but not on x or z ;
 F_z is dependent on z , but not on x or y
- The particle is displaced by $d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$
- $dW = \vec{F} \cdot d\vec{r} = F_x dx + F_y dy + F_z dz$
- Work W done by the force $\vec{F}$ while the particle moves from $\vec{r}_i$ with coordinates (x_i, y_i, z_i) to $\vec{r}_f$ with coordinates (x_f, y_f, z_f)

$$W = \int_{r_i}^{r_f} dW = \int_{x_i}^{x_f} F_x dx + \int_{y_i}^{y_f} F_y dy + \int_{z_i}^{z_f} F_z dz$$

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Sample Problem p155

- A force $\vec{F} = (3x^2 \text{ N})\hat{i} + (4 \text{ N})\hat{j}$, with x in meters, acts on a particle changing only the kinetic energy of the particle.
(a) How much work is done on the particle as it moves from coordinates (2 m, 3m) to (3 m, 0 m)?
(b) Does the speed of the particle increase, decrease or remain the same?

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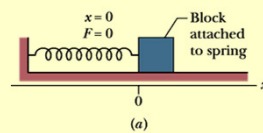
Problem 41

- A single force acts on a 3.0 kg particle-like object whose position is given by $x = 3.0t - 4.0t^2 + 1.0t^3$, with x in meters and t in seconds. Find the work done on the object by the force from $t = 0$ to $t = 4.0$ s.

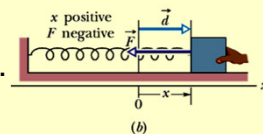
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Force due to a Spring

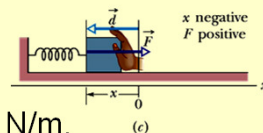
- Top figure – spring is relaxed, but attached to an object (a block)
- The spring force – **restoring force**



- If the block is pulled to the right, the spring pulls the block back to the left.
- If the block is pushed to the left, the spring will push it back to the right.



- $\vec{F} \propto \vec{d}$ $\vec{d}$ is the displacement
- $\vec{F} = -k\vec{d}$ **Hooke's Law**,
where k is the spring constant, units N/m.
- Force is in the opposite direction to the displacement.



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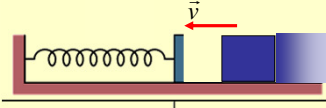
Work done by a Spring

$$\begin{aligned}
 W &= \int_{x_i}^{x_f} F(x) dx \\
 &= \int_{x_i}^{x_f} (-kx) dx = -k \int_{x_i}^{x_f} (x) dx \\
 &= -k \left[\frac{1}{2} x^2 \right]_{x_i}^{x_f} = -\frac{1}{2} k x_{xf}^2 - \left(-\frac{1}{2} k x_{xi}^2 \right) \\
 W &= \frac{1}{2} k x_{xi}^2 - \frac{1}{2} k x_{xf}^2
 \end{aligned}$$

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Sample Problem p151

- In the figure, a canister of mass $m = 0.40$ kg, slides across a horizontal frictionless surface with a speed $v = 0.50$ m/s. It then runs into and compresses a spring of spring constant $k = 750$ N/m. When the canister is momentarily stopped by the spring, by what distance is the spring compressed?



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Power

- The rate at which work is done
- $P_{avg} = \frac{W}{\Delta t}$ **Average power**
- $P = \frac{dW}{dt}$ **Instantaneous power**
- Unit: **watt (W)**
- 1 watt = 1W = 1 J/s
- Electrical consumption is often reported in kilowatt-hour
- 1kW.h = 1000 W $\times$ 3600 s = 3.6×10^6 J = 3.6 MJ

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Problem 46

- A loaded elevator has a mass of 3.0×10^3 kg and moves 210 m up the shaft in 23 s at a constant speed. At what average rate does the force from the cable do work on the elevator?

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